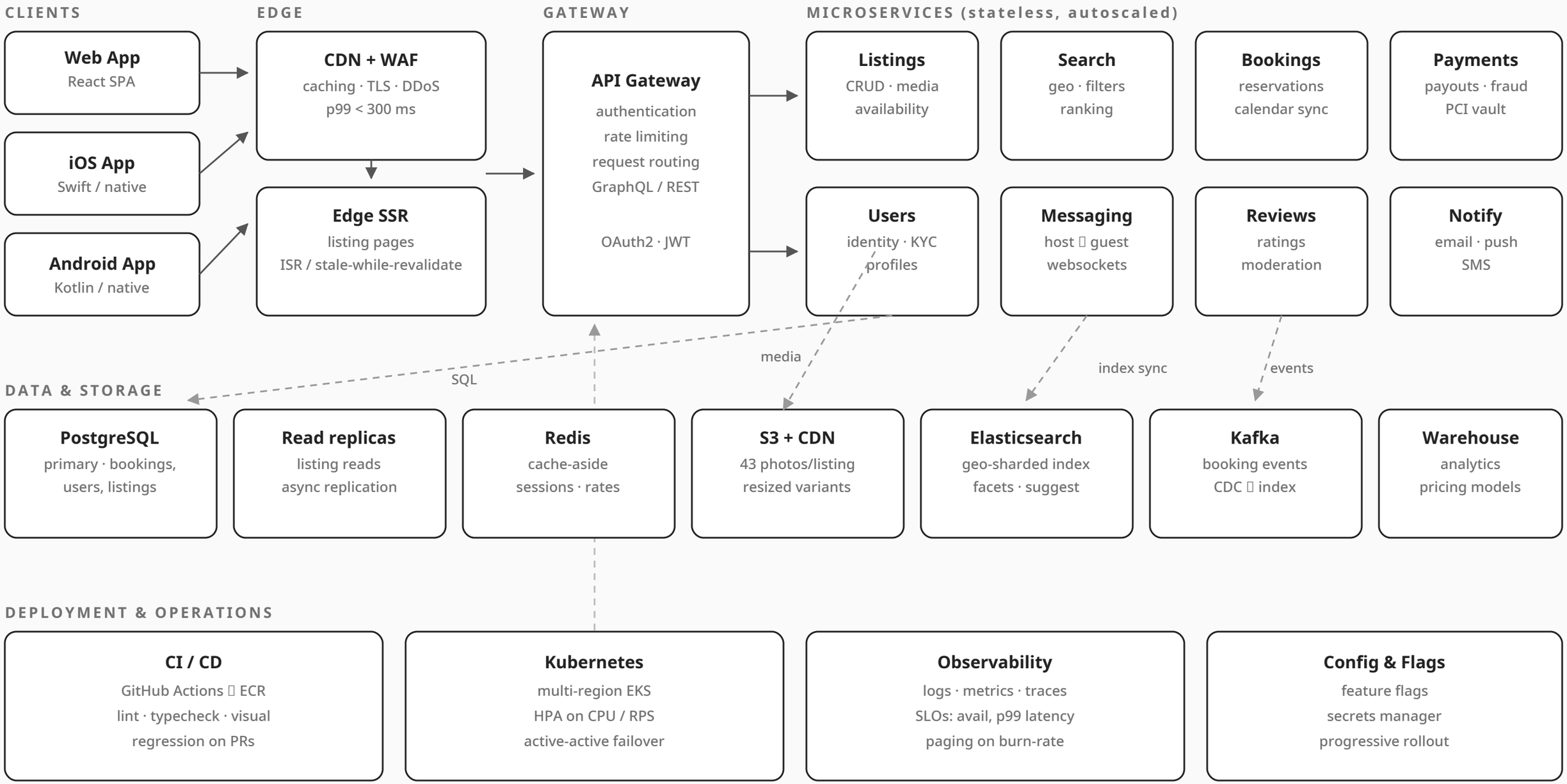
Vacation-Rental Marketplace — Production Architecture

Scaling strategy: stateless services · read replicas · cache-aside · CDN-cached SSR · async events · geo-sharded search · multi-region active-active



How it scales

- Frontend: listing pages SSR at the edge with ISR — cache hits serve from CDN PoPs; dynamic pricing/hydration via API.
- Backend: stateless microservices behind the gateway; HPA scales on RPS/CPU; bulkheads + circuit breakers per dependency.
- Storage: Postgres primary for writes + read replicas for listing reads; Redis cache-aside for hot listings; S3 stores photo variants.
- Search: Elasticsearch geo-sharded by region; Kafka CDC keeps the index fresh; ranking blends relevance, quality and conversion.
- Deployment: containers → multi-region Kubernetes; CI runs lint, typecheck and visual regression; flags enable progressive rollout; SLO-based paging.

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